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Simultaneous determination of monocrotaline and its N-oxide metabolite in rat plasma by LC–MS/MS: application to a pharmacokinetic study

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Abstract

Monocrotaline (MCT) is a pyrrolizidine alkaloid which can induce hepatic sinusoidal damage, pulmonary hypertension, renal toxicity and heart disease. Monocrotaline N-oxide (MNO), the primary metabolite of MCT, is less toxic, however it can convert back into MCT to exhibit its toxicity. In present study, a rapid and sensitivity LC-MS/MS method for simultaneous determination of MCT and MNO in rat plasma was developed and validated. The method was linearity over concentration range of 1 to 2000 ng/mL with correlation coefficients (r) > 0.997 for each analyte. The results of selectivity, matrix effect, accuracy and precision, recovery were all within the acceptance criteria. The validated method has been successfully applied to study pharmacokinetic behaviors and bioavailability of MCT in rats. The MCT was rapid absorbed (T_{max}: 0.400 ± 0.149 h) after oral administration of MCT, and the absolute bioavailability of MCT was 78.2%.

KEYWORDS: Monocrotaline, Monocrotaline N-oxide, Pharmacokinetics, LC-MS/MS

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1. INTRODUCTION

Pyrrolizidine alkaloids (PAs) are a widespread group of plant toxins (Thomas Hartmann, 1995). So far, more than 660 PAs and their corresponding N-oxides were found in more than 6000 plants over the world, mainly from four plant families: Asteraceae (or Compositae), Boraginaceae, Fabaceae (or Leguminosae), and Orchidaceae (Fu, 2017; Fu, Xia, Lin, & Chou, 2004; Li, Xia, Ruan, Fu, & Lin, 2011). Monocrotaline (MCT) is a type of toxic PA mainly found in plants of *Crotalaria* species (Lafranconi & Huxtable, 1984; Pilbeam, Lyon-Joyce, & Bell, 1983). MCT can induce multiple-organ injury, such as hepatic sinusoidal damage, pulmonary hypertension, renal toxicity and heart disease (DeLeve, Wang, Kuhlenkamp, & Kaplowitz, 1996; Imoto, Kumatani, Okada, & Yamawaki, 2016; Roth, Dotzlaf, Baranyi, Kuo, & Hook, 1981; Tanino, 2001). Monocrotaline N-oxide (MNO), the primary metabolite of MCT in vivo (Wang, Yan, Beger, Fu, & Chou, 2005), is also a toxic PA due to its conversion to MCT via intestinal and hepatic biotransformation (He, Xia, Woodling, Lin, & Fu, 2017; Yang et al., 2019).

At present, several liquid chromatography-tandem mass spectrometry (LC-MS/MS) methods were reported to determine the concentration of MCT in mice plasma (Chen et al., 2018; Yao et al., 2014). Previously, a method was reported to simultaneously determine MCT, MNO and retronecine (hydrolysis metabolite of MCT). However, the LC run time was 12 min per sample and the lower limit of quantitation (LLOQ) was 5 ng/mL for all analytes. Then Chen et. al. developed another method with lower LLOQ for MCT at 1 ng/mL and the shorter run time of 4 min. However, the method was lack of monitoring for the primary N-oxide metabolite.

Because the number of PAs with in vivo toxicity data is very limit, in vitro-to-in vivo extrapolation is an important strategy for assessment or comparison toxicity of different PAs. The differences in pharmacokinetics may be a critical factor when predicting in vivo toxicity from in vitro toxicity data. In this study, we aimed to develop a rapid, sensitive and robust LC-

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MS/MS method to simultaneously determination of MCT and its toxic N-oxide metabolite in rat plasma. The developed method was applied to investigate the pharmacokinetic profiles of MCT and MNO after intravenous or oral administration of MCT in male Sprague-Dawley rats.

2. MATERIALS AND METHODS

2.1. Chemicals and reagents

Monocrotaline (purity > 99%), monocrotaline N-oxide (purity > 98%) and erucifoline (ERU, purity > 98%) were purchased from Fluorochem Ltd. (Hadfield, UK), Chengdu Biopurify Phytochemicals Ltd. (Chengdu, China) and Phytolab GmbH & Co. (Vestenbergsgreuth, Germany), respectively. High performance liquid chromatography (HPLC)-grade acetonitrile, methanol, formic acid and ammonium acetate were obtained from Sigma-Aldrich (MO, USA). Deionized water was processed by a Milli-Q ultra-pure water purification system (Millipore, Molsheim, France).

2.2. LC-MS/MS conditions

The concentrations of MCT and MNO in rat plasma samples were quantified on a Xevo TQ-S Triple Quadrupole Mass Spectrometer with electrospray ionization (ESI) coupled with Waters Acquity^{TM} I-class system (Waters Corp, Milford, MA, USA). Chromatographic separation was performed on a Waters UPLC BEH C_{18} column (50 × 2.1 mm, 1.7 μm) at 40℃. Mobile phase A was 0.1% formic acid with 5 mM ammonium acetate in H_2O, and mobile phase B was 0.1% formic acid in acetonitrile. The optimized gradient program was used as follows: 0–1.0 min, 2% B to 40% B; 1.0–1.1 min, 40% to 95% B; 1.1–1.5 min, 95% B; 1.5–2.0 min, 2% B. The elution flow rate was set to 0.5 mL/min, and the injection volume was 2 μL.

The mass spectrometer was operated in positive ion mode. The source parameters were as follows: capillary voltage, 3.0 kV; source temperature, 150°C; desolvation temperature, 500 °C; desolvation gas flow, 1000 L/h and cone gas flow, 150 L/h. The multiple reaction monitoring

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(MRM) transitions for quantitation were *m/z* 326.1 → *m/z* 120.2 [collision energy (CE) 32 eV] for MCT, *m/z* 342.1 → *m/z* 137.0 (CE 28 eV) for MNO and *m/z* 350.1 → *m/z* 120.1 (CE 27 eV) for ERU (internal standard, IS). The cone voltage was 46 V for all the analytes. Data acquisition and processing were conducted using MassLynx 4.1 software (Waters Corp., Milford, MA, USA).

2.3. Preparation of calibrators and control samples

The stock solutions of MCT, MNO and ERU were prepared in methanol at a concentration of 1 mg/mL, respectively. The calibrators were prepared by mixing MCT and MNO in rat plasma, and serially diluted with blank rat plasma to produce concentrations range from 1 to 2000 ng/mL per analyte. The separated stock solutions of the analytes were dissolved to prepare quality control (QC) samples at four concentration levels: 1, 3, 150 and 1500 ng/mL in rat plasma, respectively. The IS working solution (ERU, 100 ng/mL) was prepared in methanol/water (1/1, v/v).

2.4. Sample preparation

An aliquot of 100 μL of MeOH/ACN (1/1, v/v) was added into the mixture containing 10-μL IS working solution (100 ng/mL) and 10-μL plasma sample to remove plasma proteins. The mixtures were mixed for 5 min and centrifuged for 5 min at 4000 rpm. An aliquot of 40-μL supernatant was mixed with 120 μL of deionized water, and a 2-μL aliquot of mixture was injected into LC-MS/MS.

2.5. Method validation

Method validation was conducted according to "Bioanalytical Method Validation Guidance for Industry" (Food and Drug Administration, 2018). The bioanalytical parameters of the method were validated included as follows: selectivity, linearity, sensitivity, carryover, accuracy, precision, recovery, matrix effect and stability in plasma samples.

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2.5.1. Selectivity and matrix effect

The selectivity of the method was the ability to differentiate the analytes from endogenous components in the blank rat plasma samples, and was evaluated by analyzing six blank samples from different individuals.

The matrix effect was investigated using blank samples from six rats at low and high QC concentration levels. The matrix factor (MF) was comparing the peak area ratio of extracted blank plasma spiked an appropriate amount of QC solution, with neat solution. The CV of IS-normalized MFs for analytes should be less than 15%.

2.5.2. Linearity, sensitivity and carryover

The concentrations of each analyte in calibration curves were determined using the peak area ratio of each analyte to IS. The calibration curves were obtained by performing a linear regression (weighting $1/x^2$) on the non-zero calibrators, and the correlation coefficient (r) should be more than 0.99. The sensitivity was defined as the lower limit of quantitation (LLOQ) in the calibration curves. The acceptance criterion was the accuracy bias of calibrators should be within ± 15.0% (± 20.0% at LLOQ) of the nominal concentration.

The carryover assay was performed by injecting a blank sample immediately following the upper limit of quantitation (ULOQ) sample in all runs.

2.5.3. Accuracy and precision

The accuracy and precision were evaluated at four QC concentration levels in rat plasma. Six replicates per QC level were analyzed in three independent runs on different days. Within-run accuracy and precision were assessed by analyzing the QCs in an individual run, and between-run accuracy and precision were assessed by analyzing the QCs in all three runs. The acceptance criteria were the accuracy bias should be within ± 15.0% (± 20.0% at LLOQ) of the nominal concentration, and the precision should not exceed 15% (20.0% at LLOQ).

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2.5.4. Recovery

The recovery was assessed at low, medium and high QC concentration levels. It was determined by comparing the peak area of analyte or IS in the QC samples extracted from rat plasma, with the reference samples prepared by spiked appropriate amount analytes and IS into blank plasma extract.

2.5.5. Stability

The stability experiments of MCT and MNO were conducted using low and high QC samples under the following storage conditions: at room temperature for 8 h; at -80℃ for 14 days; after three freeze and thaw cycles (-80℃ to room temperature); and processed samples stored in the autosampler (the temperature was set at 10℃) for 24 h. An analyte was considered stable if the mean accuracy bias was within ± 15.0% of nominal concentration at each level, and the precision should not exceed 15%.

2.5.6. Dilution integrity

The integrity of dilution was evaluated by dilution QC samples above the ULOQ with blank rat plasma to get diluted QC samples within the range of linearity. Six replicates per dilution factor were analyzed. The acceptance criteria were the mean accuracy bias should be within ± 15.0% of the nominal concentration, and the precision should not exceed 15%.

2.6. Pharmacokinetic study

Male Sprague-Dawley (SD) rats (180–220 g, 6–9 weeks) were purchased from HFK Bio-technology Co. Ltd (Beijing, China). The animal study was approved by Institutional Animal Care and Use Committee of Shanghai Institute of Materia Medica, Chinese Academy of Sciences. After fasting with access to water *ad libitum* overnight, the rats were randomly divided into two groups (n = 5 per group). MCT saline solution was intravenously administrated to rats at a dose of 1 mg/kg in group I, and rats in group II were orally

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administrated MCT saline solution at 10 mg/kg. Blood samples (~ 50 μL) were collected at 0.05, 0.167, 0.5, 1, 2, 4, 8 and 24 h after intravenous administration, while at 0.167, 0.333, 0.667, 1, 2, 4, 8 and 24 h after oral administration. The harvest plasma samples were stored at -80℃ before analysis.

3. RESULTS

3.1. Method development

The addition of internal standard (IS) is a necessity and effective method to minimize the deviations from sample preparation steps and injection volume, correct the change in mass response. In this study, ERU, a structural analogue of MCT, was selected as IS.

The MS/MS spectra of MCT, MNO and IS in the positive ESI mode were shown in Figure 1. The protonated precursor ions were selected as parent ions, and the top two abundance product ions of each analyte were monitored. Considering the highest signal to noise (S/N) ratio and the interferences in the transitions, the following MRM transitions were chosen for quantitation: *m/z* 326.1 → *m/z* 120.2 for MCT, *m/z* 342.1 → *m/z* 137.0 for MNO and *m/z* 350.1 → *m/z* 120.1 for IS.

For the chromatographic conditions, the addition of ammonium acetate can improve the S/N ratio of analytes. The elution using acetonitrile with 0.1% formic acid can obtain good peak shape and shorten the analysis time.

3.1.1. Selectivity and matrix effect

As shown in Figure 2, at the retention times of MCT (0.68 min), MNO (0.74 min) and IS (0.73 min), no interferences from the rat blank plasma were observed, respectively.

The mean IS-normalized MFs for MCT were 105% with a CV of 4.0% at low QC concentration, while 103% with a CV of 4.4% at high QC concentration. The mean IS-normalized MFs for MNO were 113% with a CV of 6.1% at low QC concentration, while 111% with a CV of 4.1%

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at high QC concentration. The results suggested there was no significant impact from the presence of rat plasma.

3.1.2. Linearity, sensitivity and carryover

The calibration curve was conducted over the range of 1–2000 ng/mL for each analyte. The eight-point calibration curve exhibited good linearity with correlation coefficient (r) greater than 0.997 for each analyte. The mean accuracy bias for MCT and MNO in three validation runs were -2.8–1.6% and -6.4–4.8%, respectively; while the CVs were less than 6.5% and 10.2%, respectively. A typical equation was $y = 0.0120x + 0.0007$ ($r^2 = 0.9984$) for MCT, and $y = 0.0113x + 0.0024$ ($r^2 = 0.9975$) for MNO, respectively. The acceptable LLOQ was 1.0 ng/mL for each analyte.

The carryover of analytes and IS were acceptable in all runs.

3.1.3. Accuracy and precision

The results of within-run and between run accuracy and precision were listed in Table 1. The results indicated the developed method is reproducible, accurate and precise for determination of MCT and MNO in rat plasma in the range of linearity.

3.1.4. Recovery

The recovery of low, medium and high QC for MCT was 90.5%, 89.6% and 85.2%, respectively, whereas for MNO was 90.5%, 89.7% and 87.0%, respectively. The mean recovery of IS was 93.4%. Recovery was found to be consistent and reproducible from rat plasma.

3.1.5. Stability

The results of stability experiments are shown in Table 2. These results suggested that rat plasma samples could be stored at room temperature for 8 h, or -80°C for up to 14 days, and the processed samples were stable after stored in the autosampler at approximate 10°C within

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24 h. After three freezing-thawing cycles (-80℃ to room temperature), the analytes were adequately stable to quantification in rat plasma.

3.1.6. Dilution integrity

The QC samples at 6000 ng/mL were diluted in blank rat plasma by factor of four into the range of linearity for both analytes. The mean accuracy bias for MCT was -7.5% with a CV of 3.4%, while -3.2% with a CV of 3.0% for MNO. The results indicated the dilution factor of four can provide precise and accurate diluted samples.

3.2. Pharmacokinetics and bioavailability

The validated analytical method was employed to quantify of MCT and MNO in rats following intravenous (IV, 1mg/kg) and oral (PO, 10 mg/kg) administration of MCT. Mean concentration-time profiles of MCT and MNO are plotted in Figure 3, and the pharmacokinetic parameters were evaluated and summarized in Table 3. Following IV administration, MCT exhibited moderate plasma clearance (1.93 ± 0.13 L/h/kg) and moderate volume of distribution (2.44 ± 0.18 L/kg). Following PO administration, MCT was rapidly absorbed (T_{max}: 0.400 ± 0.14 h) and C_{max} was 2102 ± 193 ng/mL; the C_{max} of MNO was 215 ± 37 ng/mL at 0.734 ± 0.149 h (T_{max}). In IV and PO administration rats, the terminal half-life ($t_{1/2}$) of MCT was 1.05 ± 0.12 h and 1.49 ± 0.79 h, respectively; $AUC_{0\text{-}24\ h}$ was 516 ± 34 and 4035 ± 495 ng/mL·h for MCT, respectively; $AUC_{0\text{-}24\ h}$ was 30.9 ± 4.1 and 422 ± 49 ng/mL·h for MNO, respectively. The absolute oral bioavailability was 78.2%.

The $AUC_{0\text{-}24\ h}$ ratios of MNO to MCT after oral administration (about 0.105) was approximately 1.75-fold higher than that after IV administration (about 0.060). This result indicated that in addition to metabolize in the liver, MCT may be metabolized to MNO in the intestinal microsomes and fecal microbiota.

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4. DISSCUSION

Two LC-MS/MS methods for detection of MCT in mice blood/plasma were reported (Chen et al., 2018; Yao et al., 2014). Chen et al. reported a UPLC-ESI/MS/MS method for determination of MCT in mouse blood within run time of 4 min. We applied a similar extract method but using the mixture of methanol and acetonitrile to remove the plasma protein. However, this reported method was lack of monitoring for the N-oxide metabolite, which can exert its toxicity via convert back to MCT. Another HPLC-MS/MS method was developed to quantify MCT and MNO in mice plasma, and the LC run time was 12 min per sample and the LLOQ was 5 ng/mL for both analytes. Generally, comparing our method developed in this study with the above reported methods, it demonstrated high sensitivity (LLOQ was 1 ng/mL for both analytes) and higher throughout (LC run time was 2 min per sample) to simultaneously monitor MCT and its toxic N-oxide metabolite.

5. CONCLUSION

A sensitive, fast and reliable LC-MS/MS method has been developed and validated to simultaneous determination of MCT and its N-oxide metabolite in rat plasma. The method showed a good linearity over the range of 1–2000 ng/mL for both analytes, respectively. The LLOQs were 1 ng/mL for both analytes, respectively. The total LC run time was 2.0 min. Finally, the method was applied to pharmacokinetics study of MCT in SD rats. The results showed that oral bioavailability of MCT was 78.2%。

CONFLICTS OF INTEREST

All authors declare that they have no conflicts of interest.

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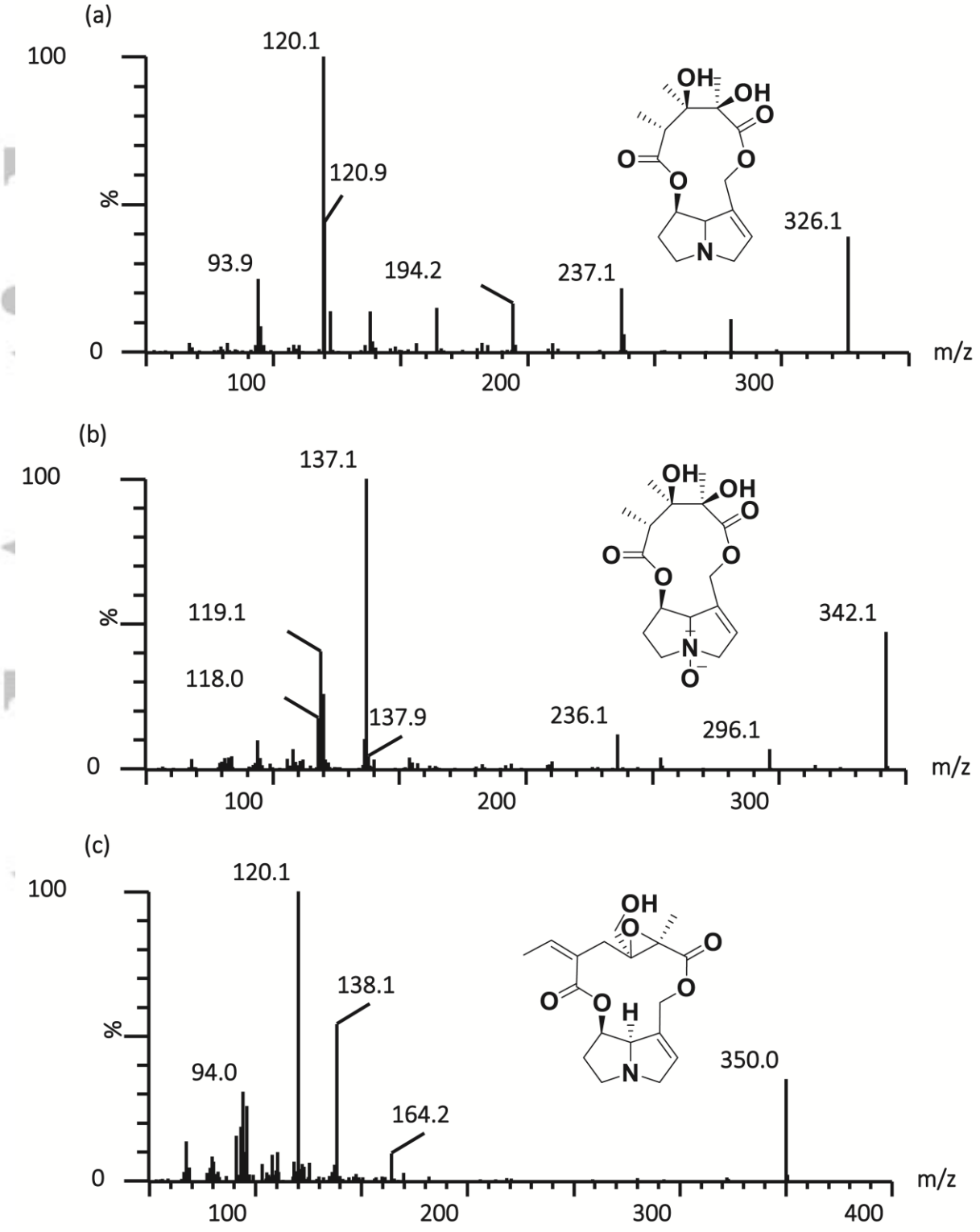


Figure 1. The MS/MS spectra of (A) MCT, (B) MNO and (C) ERU (IS).

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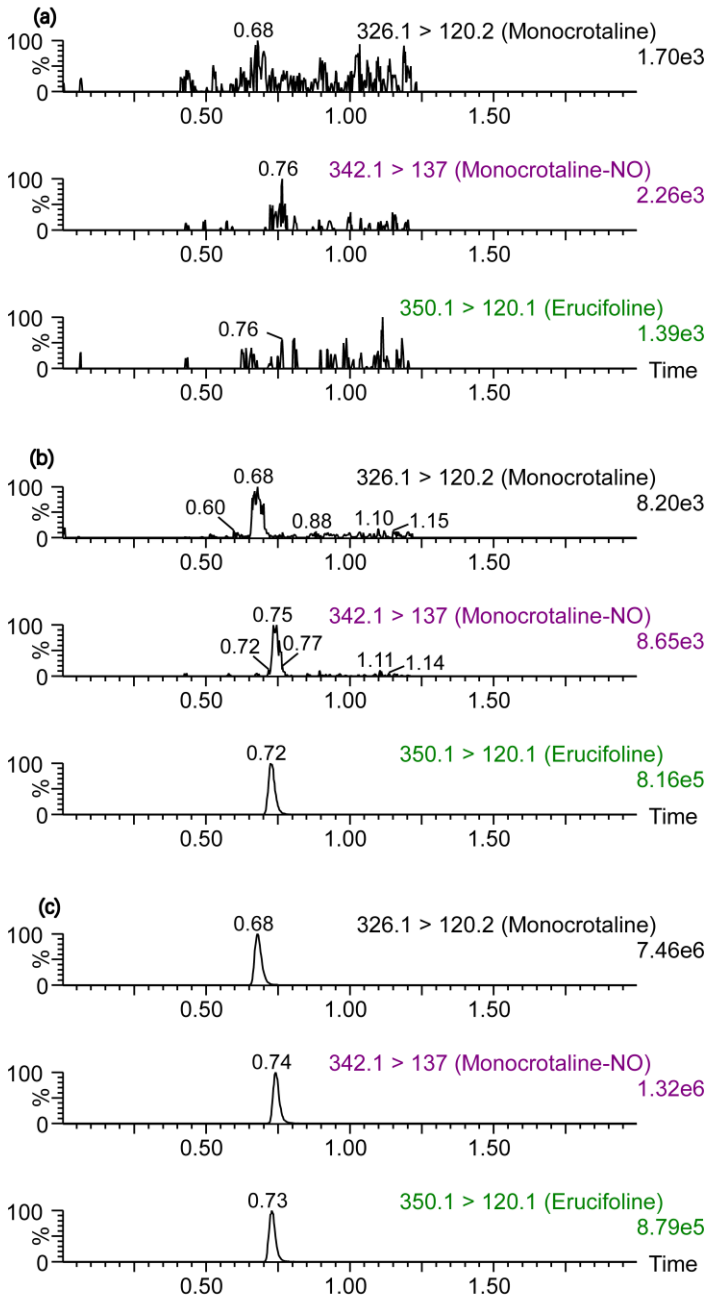


Figure 2. Representative chromatograms in rat plasma (A) blank plasma, (B) plasma spiked with analytes (1 ng/mL, LLOQ) and IS (100 ng/mL) and (C) 2 h post dose plasma sample after oral administration of 10 mg/kg MCT.

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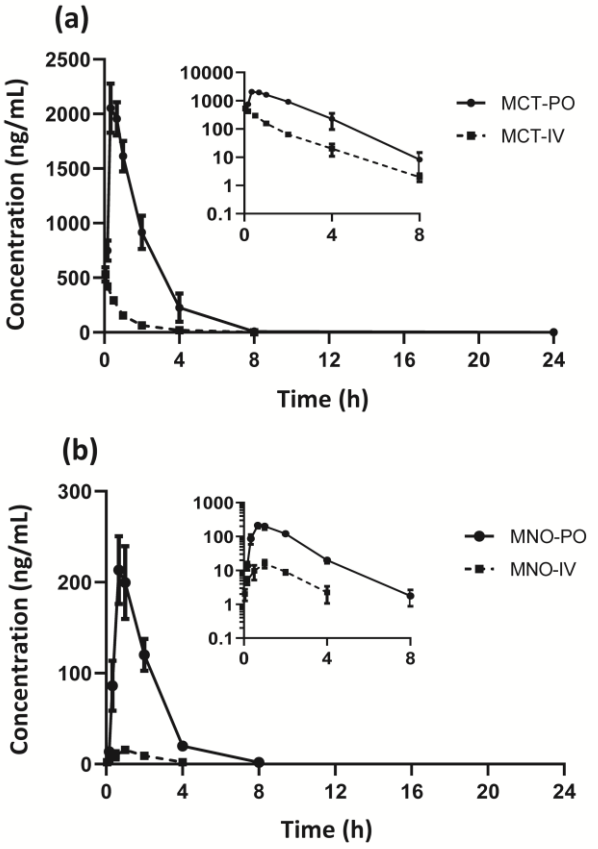


Figure 3. Mean plasma concentration-time curves of MCT and MNO in rats after IV (1 mg/kg) or PO (10 mg/kg) administration. (A) MCT; (B) MNO. The graphs are expressed as the mean ± SD, n=5.

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Table 1. Accuracy and precision for MCT and MNO in rat plasma

Analyte	Nominal Concentration (ng/mL)	Accuracy Bias (%)		Precision (%)	
		Within run	Between run	Within run	Between run
MCT	1.0	-4.0	-2.4	11.0	13.2
	3.0	3.5	2.6	5.0	6.8
	150	-1.2	-1.1	1.4	1.2
	1500	-2.5	-1.6	1.6	2.0
MNO	1.0	-6.7	-6.7	6.2	8.5
	3.0	3.0	3.1	3.8	3.2
	150	-1.1	0.0	0.6	1.2
	1500	-5.2	-3.6	1.1	2.1

Table 2. Stability of MCT and MNO in rat plasma (n=3)

Analyte	Storage Condition	Nominal Concentration (ng/mL)	Measured Concentration (ng/mL)	Accuracy Bias (%)	Precision (%)
MCT	Room temperature (8 h)	3	3.01	0.2	9.5
		1500	1494	-0.4	3.1
	-80°C (14 days)	3	3.02	0.7	10.1
		1500	1496	-0.3	1.4
	Three freeze-thaw cycle	3	3.07	2.3	5.4
		1500	1476	-1.6	1.4
	Processed sample stability[a]	3	2.94	-2.1	2.9
		1500	1398	-6.8	1.5
MNO	Room temperature (8 h)	3	3.03	1.1	2.7
		1500	1446	-3.6	1.6
	-80°C (14 days)	3	3.14	4.7	1.0
		1500	1471	-1.9	1.1
	Three freeze-thaw cycle	3	3.14	4.8	4.3
		1500	1430	-4.6	0.8
	Processed sample stability[a]	3	3.01	0.2	3.3
		1500	1448	-3.5	1.4

a. Six replicates per concentration were analyzed in the stability study of processed samples.

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Table 3. Pharmacokinetic parameters of URM and UNO in rats (mean ± SD, n = 5)

Parameters	IV (1 mg/kg)		PO (10 mg/kg)	
	MCT	MNO	MCT	MNO
T_{max} (h)	-	-	0.400 ± 0.149	0.734 ± 0.149
C_{max} (ng/mL)	531 ± 66	17.0 ± 3.1	2102 ± 193	215 ± 37
$AUC_{0\text{-}24h}$ (ng/mL·h)	516 ± 34	30.9 ± 4.1	4035 ± 495	422 ± 49
$AUC_{0\text{-}\infty}$ (ng/mL·h)	519 ± 35	35.0 ± 6.9	4045 ± 502	424 ± 49
$T_{1/2}$ (h)	1.05 ± 0.12	1.10 ± 0.43	1.49 ± 0.79	0.917 ± 0.106
$MRT_{0\text{-}24\ h}$ (h)	1.22 ± 0.13	-	1.72 ± 0.19	-
$MRT_{0\text{-}\infty}$ (h)	1.27 ± 0.13	-	1.75 ± 0.20	-
CL (L/h/kg)	1.93 ± 0.13	-	-	-
Vd (L/kg)	2.44 ± 0.18	-	-	-
%F	-	-	78.2	

Abbreviations: C_{max}, the maximum plasma concentration; T_{max}, the time to reach C_{max}; $AUC_{0\text{-}24h}$, the area under the concertation-time curve over 0–24 h; $AUC_{0\text{-}\infty}$, the area under the concertation-time curve from 0 to infinity; T1/2, the terminal half-life; $MRT_{0\text{-}24\ h}$, the mean residence time over 0–24 h; $MRT_{0\text{-}\infty}$, the mean residence time from 0 to infinity; CL: clearance; Vd, the volume of distribution; F, the absolute oral bioavailability.

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